[DISTRIBUTED SYSTEMS PROJECT]

POLYGLOT SEARCH ENGINE

CRAWL / INDEX / RANK / LEARN

PYTHON
NESTJS
VUE 3
KAFKA
POSTGRESQL
REDIS
NGINX

[THE PROBLEM]

ONE MONOLITH.
ONE BOTTLENECK.

COUPLED

Parsing and tokenizing steal capacity from live queries.

FRAGILE

A crawl burst or traffic spike degrades every layer at once.

COMPROMISED

Data-heavy indexing and business logic want different languages.

[THE ANSWER]

5

services, 2 languages, 1 gateway.

SEARCH	FastAPI · cache + rank
INDEXER	FastAPI · Kafka consumer
CRAWLER	FastAPI · polite async
AUTH	NestJS · JWT + bcrypt
ANALYTICS	NestJS · queries + CTR

[ARCHITECTURE]

THE MAP

EDGE	Vue 3 SPA and API clients enter through Nginx on :8000, routed by path.
SERVICES	Search, Indexer and Crawler on FastAPI. Auth and Analytics on NestJS.
BUS	Kafka with ZooKeeper: page_crawled, search_performed, result_clicked.
STORAGE	PostgreSQL 16 is the system of record. Redis 7 caches query results.

[STEP 01]

CRAWLER

- httpx + asyncio, fully concurrent
- 1 second minimum delay per host
- Depth-bounded, binaries filtered out
- Frontier in Postgres with SKIP LOCKED

[STEP 02]

INDEXER

- Consumes page_crawled from Kafka
- Lowercase, tokenize, drop stop-words
- Term frequency per page
- Batch upsert, ON CONFLICT DO UPDATE

[QUERY PATH]

ONE QUERY, TRACED

```
GET/search?q=database+engine
gatewayroute to search-service
redisMISS
postgresmatch terms, score by coverage + tf
snippetcontext window around matches
redisSET result for next time
kafkasearch_performed to analytics
```

[BY THE NUMBERS]

BUILT FOR SPIKES.

<50ms
CACHED QUERY

5
MICROSERVICES

3
KAFKA TOPICS

2
LANGUAGES

[THE SURFACE]

WHAT USERS TOUCH

01 VUE 3 CONSOLE

Search, crawler control, index inspector, analytics, system health, dark mode.

02 AUTH

NestJS and Passport. bcrypt hashing, stateless JWT, protected routes.

03 ANALYTICS

Top queries and click-through rate over configurable time windows.

[WHAT'S NEXT]

01 TF-IDF and BM25 ranking

02 robots.txt compliance

03 Distributed crawler workers

04 Kafka-native analytics

THANK YOU.